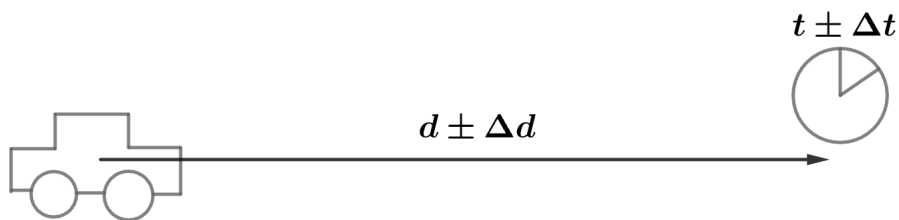


2019A F=ma Exam: Problem 11

Kevin S. Huang



We have

$$v = \frac{d}{t} = \frac{100 \text{ m}}{5.0 \text{ s}} = 20 \text{ m/s}$$

Recall when quantities are multiplied together, relative uncertainties add in quadrature so

$$\frac{\Delta v}{v} = \sqrt{\left(\frac{\Delta d}{d}\right)^2 + \left(\frac{\Delta t}{t}\right)^2}$$

Since d is known very precisely, $\Delta d = 0$:

$$\frac{\Delta v}{v} = \frac{\Delta t}{t}$$

$$\Delta v = \frac{\Delta t}{t} v = \frac{0.2 \text{ s}}{5.0 \text{ s}} (20 \text{ m/s}) = 0.8 \text{ m/s}$$

Thus, the answer is B.